

Token Optimization Package

Target: ~88% context overhead reduction

Baseline was ~72% overhead reduction (14,400 tokens saved/session). This package installs P0 glob-gating fixes and P1 protocol rules from TOKEN_OPTIMIZATION_PLAN.md.

Fastest path (recommended)

- 1. Open tools/token-optimization-package/PROMPT.md
- 2. Copy everything below the horizontal rule line
- 3. Paste into Cursor, VS Code Copilot, Windsurf, or Devin
- 4. Agent runs installer and returns Step 5 checklist

Or run two commands (repo root)

```
bash tools/token-optimization-package/install.sh --all
bash tools/token-optimization-package/verify.sh
```

P0 - Glob-gate heavy rules (done)

Rule	Fix	Tokens freed
effectiveness-signals	alwaysApply false + code globs	~1,840/session
codeguard crypto (Cursor)	alwaysApply false + globs	~500/session
codeguard certs (Cursor)	alwaysApply false + cert globs	~485/session
effectiveness (Windsurf)	alwaysApply false + globs	~1,170/session

P0 total: ~4,000 tokens/session on sessions that do not touch code files.

P1 - New protocol rules

model-routing.md

- Route LOW complexity to haiku, MEDIUM to sonnet, HIGH to opus
- Emit [ROUTING] telemetry line at task start
- ~75% cost reduction on routed low-complexity tasks

spec-length-cap.md

- spec <= 300 tokens; files_to_read <= 5 items
- No inline file contents or rule re-statements in specs
- ~20,000 tokens/month at 100 tasks

session-warm-up.md

- Read only LAST_SESSION_BRIEF.md + .comms/active/ at session start
- Never scan .comms/completed/ or full activity reports
- ~16,000 tokens/month at 20 sessions

comms-retention.md

- Cap .comms/completed/ at 20 files
- Move older files to .comms/archive/ at session end
- ~60% less directory listing noise in pipeline scans

Cumulative impact

Token Optimization - Install Instructions

Phase	Tokens/session	Monthly (20 sessions)
Baseline	14,400	288,000
+ P0 glob fixes	+4,000	+80,000
+ P1 warm-up	+800	+16,000
+ P1 spec cap (avg)	+667	+13,400
New total	~19,867	~397,400

New reduction: approximately 88% overhead vs unoptimized baseline.

What gets installed

Tool	Files	Location
Cursor	7 rules	.cursor/rules/
VS Code	7 instructions	.github/instructions/
Windsurf	5 rules	.windsurf/rules/
Devin	5 rules	.devin/rules/

Also creates .comms/archive/ and .session-logs/

P2 backlog (not installed by package)

- Compress heavy codeguard rule files to fast-path summaries
- Deduplicate .windsurf vs .cursor codeguard copies
- Cap project-progress-tracker update frequency

Implement P2 only when explicitly requested.

Works with context-handoff

Install both packages for full coverage. session-warm-up reads LAST_SESSION_BRIEF.md; context-handoff-package writes it. No conflict.

```
bash tools/context-handoff-package/install.sh --all
bash tools/token-optimization-package/install.sh --all
```

Done when

- verify.sh prints "All checks passed."
- P0 rules show alwaysApply: false with globs (Cursor/Windsurf)
- Architect returns PROMPT Step 5 checklist

Package files

- PROMPT.md - AI install prompt
- COPY_PASTE_FOR_ARCHITECT.md - email blurb
- TOKEN_OPTIMIZATION_PLAN.md - full plan
- install.sh / verify.sh / generate-pdf.sh